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Inventor(s)

Marin; Aaron et al.

ADJUSTABLE STORAGE BINS

Abstract

Adjustable storage bins for vehicles and methods for using the same are provided. An adjustable multifunction storage bin may be configured to utilize extra volume (e.g., as may be defined by or otherwise provided from any suitable B-class surface(s) (e.g., of a vehicle) or the like), which may create a variable depth/volume by allowing a user to select a floor height of the bin.

Inventors: Marin; Aaron (Lake Forest, CA), Huotari; Robert (Irvine, CA), Jenkins; Mark (Westminster, CA), Hudson; Andre (Los Angeles, CA)

Applicant: MULLEN TECHNOLOGIES, INC. (Brea, CA)

Family ID: 96881797

Assignee: MULLEN TECHNOLOGIES, INC. (Brea, CA)

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION(S) [0001] This application claims the benefit of prior filed U.S. Provisional Patent Application No. 63/559,456, filed Feb. 29, 2024, which is hereby incorporated by reference herein in its entirety.

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TECHNICAL FIELD

[0003] This disclosure relates to adjustable storage bins and, more particularly, to adjustable storage bins for vehicles and methods for using the same.

BACKGROUND OF THE DISCLOSURE

[0004] Many passenger vehicles have center consoles between adjacent seats. However, such center consoles are often limited in their storage capabilities.

SUMMARY OF THE DISCLOSURE

[0005] This document describes adjustable storage bins for vehicles and methods for using the same.

[0006] For example, a system is provided for an adjustable storage bin, as disclosed herein.

[0007] As another example, a method is provided for using an adjustable storage bin, as disclosed herein.

[0008] As yet another example, there is provided a non-transitory computer-readable storage medium storing at least one program, the at least one program including instructions, which, when executed by at least one processor, cause the at least one processor to use an adjustable storage bin, as disclosed herein.

[0009] As yet another example, there is provided a vehicle that includes a vehicle bin component defining a vehicle bin component space, and a storage bin system including a bin carrier defining a carrier space, wherein the bin carrier is at least partially positioned within the vehicle bin component space, a tray, and a tray movement assembly operative to control movement within the carrier space of the tray between a plurality of tray positions relative to the bin carrier.

[0010] As yet another example, there is provided a vehicle that includes a vehicle bin component defining a vehicle bin component space, and a storage bin system including a bin carrier defining a carrier space, wherein the bin carrier is at least partially positioned within the vehicle bin component space, a tray, and a tray movement assembly operative to move the tray within the carrier space along a first axis, and move the tray out from the carrier space along a second axis that is perpendicular to the first axis.

[0011] As yet another example, there is provided a vehicle including a vehicle bin component defining a vehicle bin component space, and a storage bin system including a bin carrier defining a carrier space, wherein the bin carrier is at least partially positioned within the vehicle bin component space, a tray, and a tray movement assembly operative to control movement within the carrier space along a first axis of the tray between a first tray position and a second tray position, wherein the bin carrier includes a wall with an opening for enabling a user to move the tray along a second axis perpendicular to the first axis from the second tray position to a third tray position external from the carrier space.

[0012] This Summary is provided to summarize some example embodiments, so as to provide a basic understanding of some aspects of the subject matter described in this document. Accordingly, it will be appreciated that the features described in this Summary are only examples and should not be construed to narrow the scope or spirit of the subject matter described herein in any way. Unless otherwise stated, features described in the context of one example may be combined or used with features described in the context of one or more other examples. Other features, aspects, and

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0013] The discussion below makes reference to the following drawings, in which like reference characters may refer to like parts throughout, and in which:

[0014] FIG. 1 is a schematic view of an illustrative system that may provide a vehicle management service in accordance with some embodiments of the disclosure;

[0015] FIG. 1A is a more detailed schematic view of a subsystem of the system of FIG. 1, in accordance with some embodiments of the disclosure;

[0016] FIG. 1B is a more detailed schematic view of a portion of the system of FIG. 1, in accordance with some embodiments of the disclosure;

[0017] FIG. 1C is a more detailed schematic view of another portion of the system of FIG. 1, in accordance with some embodiments of the disclosure;

[0018] FIG. 1D is a schematic view of a portion of a system that may provide a vehicle management service in accordance with some embodiments of the disclosure;

[0019] FIG. 2 is a top, rear, right side perspective view of a vehicle with an adjustable storage bin system, in accordance with some embodiments of the disclosure;

[0020] FIG. 2A is a top, front, left side perspective view of a portion of the vehicle of FIG. 2, in accordance with some embodiments of the disclosure;

[0021] FIG. 2B is a rear interior view of a portion of the vehicle of FIGS. 2 and 2A, taken from line IIB-IIB of FIG. 2A, in accordance with some embodiments of the disclosure;

[0022] FIG. 2C is a top, rear, left side perspective view of a portion of the vehicle of FIGS. 2, 2A, and 2B, in accordance with some embodiments of the disclosure;

[0023] FIG. 2D is a front interior view of a portion of the vehicle of FIGS. 2 and 2A-2C, taken from line IID-IID of FIG. 2A, in accordance with some embodiments of the disclosure;

[0024] FIG. 2E is a right side interior view of a portion of the vehicle of FIGS. 2 and 2A-2D, taken from line IIE-IIE of FIG. 2A, in accordance with some embodiments of the disclosure;

[0025] FIG. 2F is a rear interior view of a portion of the vehicle of FIGS. 2 and 2A-2E, taken from line IIF-IIF of FIG. 2A, in accordance with some embodiments of the disclosure;

[0026] FIG. 2G is a left side interior view of a portion of the vehicle of FIGS. 2 and 2A-2F, taken from line IIG-IIG of FIG. 2A, in accordance with some embodiments of the disclosure;

[0027] FIG. 2H is a top view of a portion of the vehicle of FIGS. 2 and 2A-2G, in accordance with some embodiments of the disclosure;

[0028] FIG. 2I is a top, rear, right side perspective view of a portion of the vehicle of FIGS. 2 and 2A-2H, in accordance with some embodiments of the disclosure;

[0029] FIG. 2J is a top, rear, left side perspective view of a portion of the vehicle of FIGS. 2 and 2A-2I, in accordance with some embodiments of the disclosure;

[0030] FIG. 2K is a left side interior view of a portion of the vehicle of FIGS. 2 and 2A-2J, taken from line IIK-IIK of FIG. 2A, in accordance with some embodiments of the disclosure;

[0031] FIG. 3 is a top, rear, left side perspective view of an adjustable storage bin system, in accordance with some embodiments of the disclosure;

[0032] FIG. 3A is a cross-sectional view of a portion of the adjustable storage bin system of FIG. 3, taken from line IIIA-IIIA of FIG. 3, in accordance with some embodiments of the disclosure;

[0033] FIG. 3B is a cross-sectional view of a portion of the adjustable storage bin system of FIGS. 3 and 3A, taken from line IIIB-IIIB of FIG. 3, in accordance with some embodiments of the disclosure;

[0034] FIG. 3C is a cross-sectional view of a portion of the adjustable storage bin system of FIGS. 3, 3A, and 3B, taken from line IIIC-IIIC of FIG. 3, in accordance with some embodiments of the disclosure;

[0035] FIG. 3D is a cross-sectional view of a portion of the adjustable storage bin system of FIGS. 3 and 3A-3C, taken from line IIID-IIID of FIG. 3, in a first configuration, in accordance with some embodiments of the disclosure;

[0036] FIG. 3D' is a cross-sectional view of a portion of the adjustable storage bin system of FIGS. 3 and 3A-3D, taken from line IIID'-IIID' of FIG. 3D, in accordance with some embodiments of the disclosure;

[0037] FIG. 3E is a cross-sectional view of a portion of the adjustable storage bin system of FIGS. 3 and 3A-3D', taken from line IIIE-IIIE of FIG. 3, in a second configuration, in accordance with some embodiments of the disclosure;

[0038] FIG. 3E' is a cross-sectional view of a portion of the adjustable storage bin system of FIGS. 3 and 3A-3E, taken from line IIIE'-IIIE' of FIG. 3E, in accordance with some embodiments of the disclosure;

[0039] FIG. 3F is a cross-sectional view of a portion of the adjustable storage bin system of FIGS. 3 and 3A-3E', taken from line IIIF-IIIF of FIG. 3, in a third configuration, in accordance with some embodiments of the disclosure;

[0040] FIG. 3F' is a cross-sectional view of a portion of the adjustable storage bin system of FIGS. 3 and 3A-3F, taken from line IIIF'-IIIF' of FIG. 3F, in accordance with some embodiments of the disclosure;

[0041] FIG. 3F'' is a cross-sectional view of a portion of the adjustable storage bin system of FIGS. 3 and 3A-3F', taken from line IIIF''-IIIF'' of FIG. 3, in various configurations, in accordance with some embodiments of the disclosure;

[0042] FIG. 3F''' is a cross-sectional view of a portion of the adjustable storage bin system of FIGS. 3 and 3A-3F'', taken from line IIIF'''-IIIF''' of FIG. 3, in various configurations, in accordance with some embodiments of the disclosure;

[0043] FIG. 3G is an exploded top, rear, left side perspective view of the adjustable storage bin system of FIGS. 3 and 3A-3F''', in accordance with some embodiments of the disclosure;

[0044] FIG. 3H is a top, rear, left side perspective view of the adjustable storage bin system of FIGS. 3 and 3A-3G, in accordance with some embodiments of the disclosure;

[0045] FIG. 3I is a top view of the adjustable storage bin system of FIGS. 3 and 3A-3H, in accordance with some embodiments of the disclosure;

[0046] FIG. 3J is a rear view of the adjustable storage bin system of FIGS. 3 and 3A-3I, in accordance with some embodiments of the disclosure;

[0047] FIG. 3K is a left side view of the adjustable storage bin system of FIGS. 3 and 3A-3J, in accordance with some embodiments of the disclosure;

[0048] FIG. 3L is a front view of the adjustable storage bin system of FIGS. 3 and 3A-3K, in accordance with some embodiments of the disclosure;

[0049] FIG. 3M is a right side view of the adjustable storage bin system of FIGS. 3 and 3A-3L, in accordance with some embodiments of the disclosure;

[0050] FIG. 3N is a bottom view of the adjustable storage bin system of FIGS. 3 and 3A-3M, in accordance with some embodiments of the disclosure;

[0051] FIG. 4 is a top, rear, left side perspective view of another adjustable storage bin system, in accordance with some embodiments of the disclosure;

[0052] FIG. 4A is a top, rear, left side perspective view of a portion of another adjustable storage bin system, in accordance with some embodiments of the disclosure;

[0053] FIG. 4' is another top, rear, left side perspective view of the adjustable storage bin system of FIG. 4, in accordance with some embodiments of the disclosure;

[0054] FIG. 4'' is a top, rear, right side perspective view of the adjustable storage bin system of

FIG. 4, in accordance with some embodiments of the disclosure;

[0055] FIG. 4' is a top, front, right side perspective view of the adjustable storage bin system of FIG. 4, in accordance with some embodiments of the disclosure;

[0056] FIG. 4'' is another top, front, right side perspective view of the adjustable storage bin system of FIG. 4, in accordance with some embodiments of the disclosure;

[0057] FIG. 4A' is a top, front, left side perspective view of the adjustable storage bin system of FIG. 4A, in accordance with some embodiments of the disclosure;

[0058] FIG. 4A'' is another top, rear, left side perspective view of the adjustable storage bin system of FIG. 4A, in accordance with some embodiments of the disclosure;

[0059] FIG. 4A''' is a top, rear, right side perspective view of the adjustable storage bin system of FIG. 4A, in accordance with some embodiments of the disclosure;

[0060] FIG. 4A'''' is another top, rear, right side perspective view of the adjustable storage bin system of FIG. 4A, in accordance with some embodiments of the disclosure;

[0061] FIG. 5 is a top, rear, left side perspective view of another adjustable storage bin system, in accordance with some embodiments of the disclosure;

[0062] FIG. 5A is an exploded top, rear, left side perspective view of the adjustable storage bin system of FIG. 5, in accordance with some embodiments of the disclosure;

[0063] FIG. 5B is a partial cross-sectional view, taken from line VB-VB of FIG. 5, of the adjustable storage bin system of FIGS. 5 and 5A, in a first configuration, in accordance with some embodiments of the disclosure; and

[0064] FIG. 5C is a partial cross-sectional view, taken from line VB-VB of FIG. 5, of the adjustable storage bin system of FIGS. 5 and 5A, in a second configuration, in accordance with some embodiments of the disclosure.

DETAILED DESCRIPTION OF THE DISCLOSURE

[0065] Adjustable storage bins for vehicles and methods for using the same are provided.

[0066] An objective is to provide extra storage opportunities and flexible areas for storage that conventionally do not include unique functions, thereby allowing users to define storage spaces that may be customized for their needs rather than fixed volume areas. An adjustable multifunction storage bin may be configured to utilize extra volume (e.g., as may be defined by or otherwise provided from any suitable B-class surface(s) (e.g., of a vehicle) or the like), which may create a variable depth/volume by allowing a user to select a floor height of the bin. To accomplish various adjusting mechanisms, this concept may be adapted to any suitable bin designs to provide maximum storage space within an available package space. An adjustable volume storage bin function may be customizable such that a bin may not be limited to storing specific items but may be reconfigurable to allow users to pursue various volume preferences for storing different types of items. Without an adjustable multifunction storage bin, a user may be stuck with a fixed bin provided by a manufacturer. For example, some packaging in in-car entertainment ("ICE") or other infotainment technologies, such as compact disc player hardware, may sacrifice B-side packaging, thereby limiting the storage areas.

[0067] FIG. 1 is a schematic view of an illustrative system 1 in which a vehicle management service, including, but not limited to adjustable multifunction storage bin management, may be facilitated amongst one or more various entities. For example, as shown in FIG. 1, system 1 may include a vehicle management service ("VMS") subsystem 10, various subsystems 100 (e.g., one or more vehicle owner ("VO") subsystems 100a-100c, one or more vehicle data collector ("VDC") subsystems 100d-100f, each of which may be communicatively coupled to one or more control modules ("CMs") 92 of a respective vehicle 90 (e.g., CMs 92a-92c of respective vehicles 90a-90c that may be owned or operated or managed or controlled by passengers or owners of respective vehicle owner subsystems 100a-100c), and at least one communications network 50 through which any two or more of the subsystems 10 and 100 may communicate. VMS subsystem 10 may be operative to interact with any of the various subsystems 100 to provide a vehicle management

service platform (“VMSP”) that may facilitate various vehicle management services, including, but not limited to, managing and enhancing the vehicle ownership process for vehicle buyers, vehicle users, and vehicle sellers for enabling effective and efficient vehicle transactions.

[0068] As shown in FIG. 1A, and as described in more detail below, a subsystem **120**, which may be any of subsystems **10**, **100**, and/or modules **92**, may include a processor component **12**, a memory component **13**, a communications component **14**, a sensor component **15**, an input/output (“I/O”) component **16**, a power supply component **17**, and/or a bus **18** that may provide one or more wired or wireless communication links or paths for transferring data and/or power to, from, or between various other components of subsystem **120**. In some embodiments, one or more components of subsystem **120** may be combined or omitted. Moreover, subsystem **120** may include other components not combined or included in FIG. 1A and/or several instances of the components shown in FIG. 1A. For the sake of simplicity, only one of each of the components of subsystem **120** is shown in FIG. 1A. I/O component **16** may include at least one input component (e.g., button, mouse, keyboard, etc.) to receive information from a user or other device or power therefrom and/or at least one output component (e.g., audio speaker, video display, haptic component, etc.) to provide information or power or any other suitable support to a user or other device, such as a touch screen that may receive input information through a user's touch of a display screen and that may also provide visual information to a user via that same display screen, or an airbag that may be deployed to provide security to one or more passengers of a vehicle user, or an adjustable multifunction storage bin, and/or the like. In some embodiments, an I/O component **16** may be any suitable data and/or power connector (e.g., a Universal Serial Bus (“USB”) connector or any other suitable connector type, a wireless charger (e.g., an inductive charging pad or the like), etc.) that may be utilized in any suitable manner by any suitable portable media device or the like. Memory **13** may include one or more storage mediums or media, including for example, a hard-drive, flash memory, permanent memory such as read-only memory (“ROM”), semi-permanent memory such as random access memory (“RAM”), any other suitable type of storage component, or any combination thereof (e.g., for storing any suitable data (e.g., data **19d**)). Communications component **14** may be provided to allow subsystem **120** to communicate with one or more other subsystems **120** (e.g., any communication to/from/between subsystem(s)/module(s) **10**, **100**, and **92** of system **1**) using any suitable communications protocol. Communications component **14** can be operative to create or connect to a communication network or link of a network. Communications component **14** can provide wireless communications using any suitable short-range or long-range communications protocol, such as Wi-Fi (e.g., an 802.11 protocol), Bluetooth, ultra-wideband, radio frequency systems (e.g., 1200 MHz, 2.4 GHz, and 5.6 GHz communication systems), near field communication (“NFC”), infrared, protocols used by wireless and cellular telephones and personal e-mail devices, or any other protocol supporting wireless communications. Communications component **14** can also be operative to connect to a wired communications link or directly to another data source wirelessly or via one or more wired connections or other suitable connection type(s). Communications component **14** may be a network interface that may include the mechanical, electrical, and/or signaling circuitry for communicating data over physical links that may be coupled to other devices of a network. Such network interface(s) may be configured to transmit and/or receive any suitable data using a variety of different communication protocols, including, but not limited to, TCP/IP, UDP, ATM, synchronous optical networks (“SONET”), any suitable wired protocols or wireless protocols now known or to be discovered, Frame Relay, Ethernet, Fiber Distributed Data Interface (“FDDI”), and/or the like. In some embodiments, one, some, or each of such network interfaces may be configured to implement one or more virtual network interfaces, such as for Virtual Private Network (“VPN”) access.

[0069] Sensor **15** may be any suitable sensor that may be configured to sense any suitable data for subsystem **120** (e.g., location-based data via a GPS sensor system, motion data, environmental data, biometric data, etc.). Sensor **15** may be a sensor assembly that may include any suitable

sensor or any suitable combination of sensors operative to detect movements of subsystem **120** and/or of any user thereof and/or any other characteristics of subsystem **120** and/or of its environment (e.g., physical activity or other characteristics of a user of subsystem **120**, light content of the device environment, gas pollution content of the device environment, noise pollution content of the device environment, altitude of the device, etc.). Sensor **15** may include any suitable sensor(s), including, but not limited to, one or more of a GPS sensor, wireless communication sensor, accelerometer, directional sensor (e.g., compass), gyroscope, motion sensor, pedometer, passive infrared sensor, ultrasonic sensor, microwave sensor, a tomographic motion detector, a camera, a biometric sensor, a light sensor, a timer, or the like. Sensor **15** may include any suitable sensor components or subassemblies for detecting any suitable movement of subsystem **120** and/or of a user thereof. For example, sensor **15** may include one or more three-axis acceleration motion sensors (e.g., an accelerometer) that may be operative to detect linear acceleration in three directions (i.e., the x- or left/right direction, the y- or up/down direction, and the z- or forward/backward direction). As another example, sensor **15** may include one or more single-axis or two-axis acceleration motion sensors that may be operative to detect linear acceleration only along each of the x- or left/right direction and the y- or up/down direction, or along any other pair of directions. In some embodiments, sensor **15** may include an electrostatic capacitance (e.g., capacitance-coupling) accelerometer that may be based on silicon micro-machined micro electro-mechanical systems (“MEMS”) technology, including a heat-based MEMS type accelerometer, a piezoelectric type accelerometer, a piezo-resistance type accelerometer, and/or any other suitable accelerometer (e.g., which may provide a pedometer or other suitable function). Sensor **15** may be operative to directly or indirectly detect rotation, rotational movement, angular displacement, tilt, position, orientation, motion along a non-linear (e.g., arcuate) path, or any other non-linear motions. Additionally or alternatively, sensor **15** may include one or more angular rate, inertial, and/or gyro-motion sensors or gyroscopes for detecting rotational movement. For example, sensor **15** may include one or more rotating or vibrating elements, optical gyroscopes, vibrating gyroscopes, gas rate gyroscopes, ring gyroscopes, magnetometers (e.g., scalar or vector magnetometers), compasses, and/or the like. Any other suitable sensors may also or alternatively be provided by sensor **15** for detecting motion on subsystem **120**, such as any suitable pressure sensors, altimeters, or the like. Using sensor **15**, subsystem **120** may be configured to determine a velocity, acceleration, orientation, and/or any other suitable motion attribute of subsystem **120** (e.g., a direction and/or strength of an impact (e.g., a crash involving a vehicle **90**)). One or more biometric sensors may be multi-modal biometric sensors and/or operative to detect long-lived biometrics, modern liveness (e.g., active, passive, etc.) biometric detection, and/or the like. Sensor **15** may include a microphone, camera, scanner (e.g., a barcode scanner or any other suitable scanner that may obtain product identifying information from a code, such as a linear barcode, a matrix barcode (e.g., a quick response (“QR”) code), or the like), proximity sensor, light detector, temperature sensor, motion sensor, biometric sensor (e.g., a fingerprint reader or other feature (e.g., facial) recognition sensor, which may operate in conjunction with a feature-processing application that may be accessible to subsystem **120** for attempting to authenticate a user), line-in connector for data and/or power, and/or combinations thereof. In some examples, each sensor can be a separate device, while, in other examples, any combination of two or more of the sensors can be included within a single subsystem or device. For example, a gyroscope, accelerometer, photoplethysmogram, galvanic skin response sensor, and temperature sensor can be included within a wearable electronic device, such as a smart watch, while a scale, blood pressure cuff, blood glucose monitor, SpO2 sensor, respiration sensor, posture sensor, stress sensor, and asthma inhaler can each be separate devices. While specific examples are provided, it should be appreciated that other sensors can be used and other combinations of sensors can be combined into a single subsystem or device. Subsystem **120** can further include a timer that can be used, for example, to add time dimensions to various attributes of any detected element(s). Sensor **15** may include any

suitable sensor components or subassemblies for detecting any suitable characteristics of any suitable condition of the lighting of the environment of subsystem **120**. For example, sensor **15** may include any suitable light sensor that may include, but is not limited to, one or more ambient visible light color sensors, illuminance ambient light level sensors, ultraviolet (“UV”) index and/or UV radiation ambient light sensors, and/or the like. Any suitable light sensor or combination of light sensors may be provided for determining the illuminance or light level of ambient light in the environment of subsystem **120** (e.g., in lux or lumens per square meter, etc.) and/or for determining the ambient color or white point chromaticity of ambient light in the environment of subsystem **120** (e.g., in hue and colorfulness or in x/y parameters with respect to an x-y chromaticity space, etc.) and/or for determining the UV index or UV radiation in the environment of subsystem **120** (e.g., in UV index units, etc.). Sensor **15** may include any suitable sensor components or subassemblies for detecting any suitable characteristics of any suitable condition of the air quality of the environment of subsystem **120**. For example, sensor **15** may include any suitable air quality sensor that may include, but is not limited to, one or more ambient air flow or air velocity meters, ambient oxygen level sensors, volatile organic compound (“VOC”) sensors, ambient humidity sensors, ambient temperature sensors, and/or the like. Any suitable ambient air sensor or combination of ambient air sensors may be provided for determining the oxygen level of the ambient air in the environment of subsystem **120** (e.g., in O₂ % per liter, etc.) and/or for determining the air velocity of the ambient air in the environment of subsystem **120** (e.g., in kilograms per second, etc.) and/or for determining the level of any suitable harmful gas or potentially harmful substance (e.g., VOC (e.g., any suitable harmful gasses, scents, odors, etc.) or particulate or dust or pollen or mold or the like) of the ambient air in the environment of subsystem **120** (e.g., in HG % per liter, etc.) and/or for determining the humidity of the ambient air in the environment of subsystem **120** (e.g., in grams of water per cubic meter, etc. (e.g., using a hygrometer)) and/or for determining the temperature of the ambient air in the environment of subsystem **120** (e.g., in degrees Celsius, etc. (e.g., using a thermometer)). Sensor **15** may include any suitable sensor components or subassemblies for detecting any suitable characteristics of any suitable condition of the sound quality of the environment of subsystem **120**. For example, sensor **15** may include any suitable sound quality sensor that may include, but is not limited to, one or more microphones or the like that may determine the level of sound pollution or noise in the environment of subsystem **120** (e.g., in decibels, etc.). Sensor **15** may also include any other suitable sensor for determining any other suitable characteristics about a user of subsystem **120** and/or the environment of subsystem **120** and/or any situation within which subsystem **120** may exist. For example, any suitable clock and/or position sensor(s) may be provided to determine the current time and/or time zone within which subsystem **120** may be located. Sensor **15** may be embedded in a body (e.g., housing **11**) of subsystem **120**, such as along a bottom surface that may be operative to contact a user, or can be positioned at any other desirable location. In some examples, different sensors can be placed in different locations inside or on the surfaces of subsystem **120** (e.g., some located inside housing **11** (e.g., any suitable component of a vehicle) and some attached to an attachment mechanism (e.g., a wrist band coupled to a housing of a wearable device), or the like). In other examples, one or more sensors can be worn by a user separately as different parts of a single subsystem **120** or as different subsystems or devices. In such cases, the sensors can be configured to communicate with subsystem **120** using a wired and/or wireless technology (e.g., via communications component **14**). In some examples, sensors can be configured to communicate with each other and/or share data collected from one or more sensors.

[0070] Power supply **17** can include any suitable circuitry for receiving and/or generating power, and for providing such power to one or more of the other components of subsystem **120**. For example, power supply assembly **17** can be coupled to a power grid (e.g., when subsystem **120** is not acting as a portable device or when a battery of the device is being charged at an electrical outlet with power generated by an electrical power plant). As another example, power supply

assembly **17** may be configured to generate power from a natural source (e.g., solar power using solar cells). As another example, power supply assembly **17** can include one or more batteries for providing power (e.g., when subsystem **120** is acting as a portable device). Subsystem **120** may also be provided with a housing **11** that may at least partially enclose one or more of the components of subsystem **120** for protection from debris and other degrading forces external to subsystem **120**. Each component of subsystem **120** may be included in the same housing **11** (e.g., as a single unitary device, such as a portable media device or server) and/or different components may be provided in different housings (e.g., a keyboard input component may be provided in a first housing that may be communicatively coupled to a processor component and a display output component that may be provided in a second housing, such as in a desktop computer set-up). In some embodiments, subsystem **120** may include other components not combined or included in those shown or several instances of the components shown.

[0071] Processor **12** may be used to run one or more applications, such as an application **19** that may be accessible from memory **13** (e.g., as a portion of data **19d**) and/or any other suitable source (e.g., from any other device in its system). Application **19** may include, but is not limited to, one or more operating system applications, firmware applications, communication applications (e.g., for enabling communication of data between devices), third party service applications, internet browsing applications (e.g., for interacting with a website provided by a third party subsystem), application programming interfaces (“APIs”), software development kits (“SDKs”), proprietary applications (e.g., a web application or a native application) for enabling subsystem **120** to interact with an online service and/or one or more other subsystems and/or the like, which may include applications for routing protocols, SDN modules based on OpenFlow, P4, or other network data plane programming standards, machine learning algorithms, network management functions, etc., any other suitable applications, such as applications for detecting and reacting to impact on a vehicle and/or detecting and reacting to positioning of travelers and components (e.g., seats, sun roof, etc.) of and within a vehicle (e.g., to adjust the position of any vehicle feature (e.g., sunroof or seat or passenger or adjustable multifunction storage bin) and/or deploy an airbag or the like), applications for detecting and reacting to communicative coupling to/decoupling from any suitable portable media device, and/or the like. For example, processor **12** may load an application **19** as an interface program to determine how instructions or data received via an input component of I/O component **16** or other component of subsystem **120** (e.g., sensor **15** and/or communications component **14**) may manipulate the way in which information may be stored (e.g., in memory **13**) and/or provided via an output component of I/O component **16** and/or communicated to another system device via communications component **14**. As one example, application **19** may be a third party application that may be running on subsystem **120** (e.g., an application associated with the network of system **1**) that may be loaded on subsystem **120** in any suitable manner, such as via an application market (e.g., using communications component **14**), such as the Apple App Store or Google Play, or that may be accessed via an internet application or web browser (e.g., by Apple Safari or Google Chrome) that may be running on subsystem **120** and that may be pointed to a uniform resource locator (“URL”) whose target or web resource may be managed by or otherwise affiliated with any suitable entity. Any subsystem may include any suitable special purpose hardware (e.g., hardware support of high-speed packet processing, hardware support of machine learning algorithms, etc.).

[0072] Subsystem **120** may be any portable, mobile, wearable, implantable, or hand-held electronic device configured to operate with system **1**. Alternatively, subsystem **120** may not be portable during use, but may instead be generally stationary (e.g., permanently coupled to a vehicle). Subsystem **120** can include, but is not limited to, a media player, video player, still image player, game player, other media player, music recorder, movie or video camera or recorder, still camera, other media recorder, radio, medical equipment, domestic appliance, smart appliance (e.g., smart door knob, smart door lock, etc.), transportation vehicle instrument, musical instrument, calculator,

cellular telephone, other wireless communication device, personal digital assistant, remote control, pager, computer (e.g., a desktop, laptop, tablet, server, etc.), monitor, television, stereo equipment, set up box, set-top box, wearable device, boom box, modem, router, printer, kiosk, beacon, server, adjustable multifunction storage bin, and any combinations thereof.

[0073] In some embodiments, processor **12** may be used to run one or more applications that may be accessible from memory **13** and/or from any other suitable source (e.g., an application from VMS subsystem **10** via an active internet connection or otherwise at and for use by a subsystem **100**). Such an application may include, but is not limited to, one or more operating system applications, firmware applications, communication applications, internet browsing applications (e.g., for interacting with a website provided by VMS subsystem **10** for enabling a subsystem **100** to interact with an online service of VMS subsystem **10** (e.g., a VMSP)), VMS applications (e.g., a web application or a native application or a hybrid application that may be at least partially produced by VMS subsystem **10** for enabling a subsystem **100** to interact with an online service of VMS subsystem **10** (e.g., a VMSP)), or any other suitable applications. As one example, an application of a subsystem **100** may provide a user or a communicatively coupled device (e.g., control module **92**) with the ability to interact with a vehicle management service or the VMSP of VMS subsystem **10**, where such an application may be a third party application that may be running on a subsystem **100** (e.g., an application (e.g., software and/or firmware) associated with VMS subsystem **10** that may be loaded on subsystem **100** from VMS subsystem **10** or via an application market) and/or that may be accessed via an internet application or web browser running on subsystem **100** (e.g., processor **12**) that may be pointed to a uniform resource locator (“URL”) whose target or web resource may be managed by VMS subsystem **10** or any other remote subsystem. Each subsystem **100** may be a portable media device (e.g., a smartphone), a laptop computer, a tablet computer, a desktop computer, an appliance, a wearable electronic device, a virtual reality device, a dongle device, at least one web or network server (e.g., for providing an online resource, such as a website or native online application, for presentation on one or more other subsystems) with an interface for an administrator of such a server, and/or the like.

[0074] Some or all portions of VMS subsystem **10** may be operated, managed, or otherwise at least partially controlled by an entity (e.g., administrator) responsible for providing a vehicle management service to one or more clients or other suitable entities. VMS subsystem **10** may communicate with one or more subsystems **100** via communications network **50**. Network **50** may be the internet or any other suitable network, such that when intercoupled via network **50**, any two subsystems of system **1** may be operative to communicate with one another (e.g., a subsystem **100** may access information (e.g., from an application **19** or data **19d** of VMS subsystem **10**, as may be provided as a vehicle management service via processor **12** and communications component **14** of VMS subsystem **10**) as if such information were stored locally at that subsystem **100** (e.g., in its memory component **13**)).

[0075] Various clients and/or partners may be enabled to interact with VMS subsystem **10** for enabling the vehicle management services and the VMSP. For example, at least one vehicle owner subsystem of system **1** (e.g., each one of the one or more vehicle owner subsystems **100a-100c**) may be any suitable subsystem (e.g., portable computer) operated by any suitable vehicle owner (“VO”) that may own, rent, or otherwise have access to a vehicle (e.g., a respective one of the one or more vehicles **90a-90c** (e.g., any suitable motor vehicle (e.g., car, truck, bus, motorcycle, etc. (e.g., a road vehicle))), railed vehicle (e.g., train, tram, etc.), watercraft (e.g., ship, boat, jet ski, etc.), aircraft (e.g., airplane, helicopter, drone, etc.), spacecraft, and/or the like)). At least one vehicle data collector subsystem of system **1** (e.g., each one of the one or more vehicle data collector subsystems **100d-100f**) may be any suitable subsystem (e.g., dongle device) that may be communicatively coupled to a respective vehicle owner subsystem (e.g., via a network **50**) and to a respective control module (e.g., via direct installation) of a respective vehicle (e.g., VDC subsystem **100d** may be communicatively coupled to VO subsystem **100a** and to CM **92a** of

vehicle **90a** that may be owned by the operator of VO subsystem **100a**, VDC subsystem **100e** may be communicatively coupled to VO subsystem **100b** and to CM **92b** of vehicle **90b** that may be owned by the operator of VO subsystem **100b**, and VDC subsystem **100f** may be communicatively coupled to VO subsystem **100c** and to CM **92c** of vehicle **90c** that may be owned by the operator of VO subsystem **100c**). For example, a VDC subsystem may be any suitable on-board diagnostics (“OBD”) device that may be operative to be communicatively coupled with any suitable control module of any suitable vehicle (e.g., via any suitable OBD-II data link connector of a vehicle (e.g., via a physical connection or wireless path)) that may be operative to monitor any suitable data from an engine control unit and/or electronic control unit (“ECU”) of the vehicle and/or from any other data source of the vehicle that may be made available (e.g., according to the OBD protocol), such as a powertrain control module (“PCM”) or otherwise. A VDC subsystem may be operative to send one or more requests to the CM of a vehicle for one or more specific parameters using one or more specific parameter identification numbers (“PIDs”) (e.g., according to the Society of Automotive Engineers (“SAE”) standard J1979) and then the VDC subsystem may communicate any received parameter data from the vehicle to a VO subsystem that may be communicatively coupled to the VDC subsystem (e.g., via any suitable wired or wireless communication protocol). For example, as shown in FIG. 1B, VDC subsystem **100d** may be communicatively coupled to any suitable control module connector **93a** via any suitable communications path **55a**, which may be a direct physical connection between connector **93a** and a connector of VDC subsystem **100d** (e.g., a male connector of an I/O component **16** of VDC subsystem **100d** may physically mate with a female control module connector **93a** (e.g., any suitable OBD-II data link connector)), where control module connector **93a** may be communicatively coupled to one, some, or all suitable control modules or data sources (e.g., control module **92a**) of vehicle **90a**, while VDC subsystem **100d** may be communicatively coupled to VO subsystem **100a** via any suitable communications path **55b** (e.g., any suitable wired or wireless communications path using any suitable communications protocol (e.g., Bluetooth between a communications component **14** of VDC subsystem **100d** and a communications component **14** of VO subsystem **100a**), while VO subsystem **100a** may be communicatively coupled to VMS subsystem **10** via any suitable communications path **55c** (e.g., any suitable wired or wireless communications path (e.g., of network **50** of FIG. 1) using any suitable communications protocol). Alternatively or additionally, as shown in FIG. 1B, VDC subsystem **100d** may be communicatively coupled to VMS subsystem **10** via any suitable communications path **55d** (e.g., any suitable wired or wireless communications path (e.g., of network **50** of FIG. 1) using any suitable communications protocol (e.g., any suitable long-range communications protocol between a communications component **14** of VDC subsystem **100d** and a communications component **14** of VMS subsystem **10** (e.g., using a low power communications component and/or any suitable telemetry functionality)) without VO subsystem **100a** as an intermediary. Additionally or alternatively, in some embodiments, a VO subsystem may be configured to communicate directly with a CM of a vehicle without the need for a distinct intermediary VDC subsystem. For example, as shown in FIG. 1C, VO subsystem **100b** may be communicatively coupled to any suitable control module connector **93b** via any suitable communications path **55e**, which may be a direct wired connection between connector **93b** and a connector of VO subsystem **100b** (e.g., a connector of an I/O component **116** of VO subsystem **100b** may be communicatively coupled to a first connector of a cable of communications path **55e** and a second connector of such a cable may be communicatively coupled with control module connector **93b** (e.g., any suitable OBD-II data link connector)), where control module connector **93b** may be communicatively coupled to one, some, or all suitable control modules or data sources (e.g., control module **92b**) of vehicle **90b**, while VO subsystem **100b** may be communicatively coupled to VMS subsystem **10** via any suitable communications path **55f** (e.g., any suitable wired or wireless communications path (e.g., of network **50** of FIG. 1) using any suitable communications protocol). In some embodiments, communications path **55e** may be a wireless

communications path between control module **92b** and VO subsystem **100b** (e.g., a wireless (e.g., Bluetooth) communication path between a communications component **14** of VO subsystem **100b** and a communications component of control module **92b** of vehicle **90b**), such that a data connection may be facilitated directly between a user's portable electronic device and a computer of a vehicle directly through a wireless connection.

[0076] Each subsystem **100** of system **1** (e.g., each one of subsystems **100a-100f**) may be operated by any suitable entity for interacting in any suitable way with VMS subsystem **10** (e.g., via network **50**) for deriving value from and/or adding value to a service of the VMSP of VMS subsystem **10**.

For example, a particular subsystem **100** may be a server operated by a client/partner entity that may receive any suitable data from VMS subsystem **10** related to any suitable vehicle management enhancement of the VMSP provided by VMS subsystem **10** (e.g., via network **50**). Additionally or alternatively, a particular subsystem **100** may be a server operated by a client/partner entity that may upload or otherwise provide any suitable data to VMS subsystem **10** related to any suitable vehicle management service of the VMSP provided by VMS subsystem **10** (e.g., via network **50**).

[0077] FIGS. **2** and **2A-2K** show an illustrative vehicle **290** that may include at least one adjustable storage bin system **250**, in accordance with some embodiments of the disclosure. Vehicle **290** may be any suitable vehicle, which may be similar to any vehicle **90** (e.g., vehicles **90a-90c**) described herein. As shown, in some embodiments, vehicle **290** may be any suitable automobile with at least one seat, which may include at least a front main seat **214fm** (e.g., a driver seat if the vehicle is able to be driven) for a front main passenger (not shown). Additionally, in some embodiments, as shown, vehicle **290** may also include a front auxiliary seat **214fa** for a front auxiliary passenger (not shown), and at least one rear seat for one or more rear passengers, such as a rear main seat **214rm** for a rear main passenger (not shown) and/or a rear auxiliary seat **214ra** for a rear auxiliary passenger (not shown). Although each seat for each passenger may be distinct in one or more ways from one another (see, e.g., distinct seats **214rm** and **214ra** of FIGS. **2** and **2A**), any vehicle may be provided with a single seat that may be used simultaneously by two or more adjacent passengers (e.g., a bench type rear seat). As also shown, vehicle **290** may include at least one vehicle subsystem **220f** (e.g., electronic vehicle subsystem) similar to any subsystem **120** that may provide a screen user interface for providing any suitable information to one or more passengers of the vehicle (e.g., a screen that may be permanently coupled to vehicle **290** (e.g., to a dashboard **230d** or otherwise), such as an infotainment screen, which may be facing or otherwise accessible not only to any passenger(s) of front seat(s) **214fm** and/or **214fa** but also to any passenger(s) of rear seat(s) **214rm** and/or **214ra**). Additionally or alternatively, vehicle may include a rear vehicle subsystem **220r** similar to any subsystem **120** that may provide a screen user interface for providing any suitable information to one or more passengers of the vehicle (e.g., a screen that may be permanently coupled to vehicle **290** (e.g., to a rear portion of a console **240c** (e.g., a floor console, a center floor console, a center console, a storage console, a first row console, a front console, etc.) or otherwise), such as an infotainment screen, which may be facing or otherwise accessible to passenger(s) of rear seat(s) **214rm** and/or **214ra**).

[0078] An adjustable storage bin system may be incorporated into or otherwise provided by or coupled to any suitable portion of any suitable vehicle. For example, as shown, adjustable storage bin system **250** may be incorporated into or otherwise provided by or coupled to console **240c**, which may extend rearward from dashboard **230d** and between at least a portion of front seats **214fm** and/or **214fa** and/or between at least a portion of rear seats **214rm** and/or **214ra**. As shown, console **240c** may include one or more cup holders, such as a front set of cup holders **240hf** (e.g., with a cup **240p** (see, e.g., FIG. **2K**)) and/or a rear set of cup holders **240hr**. Additionally or alternatively, as shown, one or more vehicle subsystems, such as subsystem **220f** and/or subsystem **220r** may be coupled to and/or extend from console **240c**. As a particular example, as shown, adjustable storage bin system **250** (e.g., adjustable storage bin system **250a**) may be positioned along console **240c** just in front of front set of cupholders **240hf** and just rear of dashboard **230d**.

Although, in other embodiments, an adjustable storage bin system **250b** may be positioned along console **240c** just rear of front set of cupholders **240hf** (e.g., for use by front passenger(s)) and/or an adjustable storage bin system **250c** may be positioned along console **240c** just in front of rear set of cupholders **240hr** and/or an adjustable storage bin system **250d** may be positioned along console **240c** just rear of rear set of cupholders **240hr** and/or an adjustable storage bin system **250e** may be positioned along console **240c** just rear of rear vehicle subsystem **220r** and/or an adjustable storage bin system **250f** may be positioned along a rear portion console **240c** (e.g., for use by rear passenger(s) (e.g., a rear center console or second row console, etc.)). An adjustable storage bin system may be positioned in any other suitable location or in any other suitable vehicle bin component in addition to or as an alternative to a center console (e.g., console **240c**), including, but not limited to, a glove box bin or interior storage bin (e.g., a bin **250g** at a glove box **249g** of vehicle **290**), a door trim bin (e.g., a bin **250h** at a door trim portion **249h** of any suitable door of vehicle **290**), an overhead storage bin (e.g., a bin **250i** at an overhead portion **249i** of any suitable roof of vehicle **290**), a trunk trim quarter garnish bin (e.g., a bin **250j** at any suitable trunk trim garnish **249j** (e.g., quarter garnish) of any suitable trunk portion of vehicle **290**), a floor panel bin (e.g., a bin **250k** at any suitable floor panel portion **249k** of any suitable floor portion (e.g., in front of any suitable seat) of vehicle **290**), a seat bin (e.g., a bin **250l** at any suitable seat portion **249l** of any suitable seat of vehicle **290**), an upper panel or an instrument panel storage bin (e.g., a bin **250m** at any suitable upper panel or instrument panel **249m** of vehicle **290**), and/or the like (e.g., any space with enough depth to adequately support a bin may be used). A vehicle bin component that may host an adjustable storage bin system may be positioned within a passenger host space **290us** in which one or more passengers may be located during use of vehicle **290** (e.g., passenger host space **290us** may be defined by the interior of the floor, ceiling, doors, pillars, windows, and/or the like of the vehicle, while the vehicle bin component may be accessible to a passenger within host space **290us** for utilizing an adjustable storage bin system positioned within that vehicle bin component (e.g., a console bin, glove box bin, door trim bin, overhead storage bin, floor panel bin, seat bin, upper panel bin, etc.)). In other embodiments, a vehicle bin component that may host an adjustable storage bin system may be positioned outside of passenger host space **290us**, whereby the vehicle bin component may not be accessible to a passenger within host space **290us**, such that the passenger may access the adjustable storage bin system when outside host space **290us** (e.g., a trunk trim garnish bin or front hood trim garnish bin).

[0079] As shown in FIGS. 2I-2K, console **240c** may include one or more openings **240co** for providing access to storage bin **250** (e.g., storage bin **250a**) that may be positioned within a vehicle bin component space or console space **241s** that may be defined by interior surface(s) of any suitable vehicle bin component walls **240cw** or other portions of console **240c** or of any other suitable vehicle bin component (e.g., glove box bin, door trim bin, overhead storage bin, trunk trim garnish bin, floor panel bin, seat bin, upper panel bin, etc.) of vehicle **290** (e.g., vehicle host space of a vehicle host component). For example, storage bin **250** may be positioned within space **241s** to take up a bin space portion **241sb** of console space **241s** while a remaining space portion **241sr** of console space **241s** may be used for other purposes, while a first console opening **240com** may be provided through a top wall **240ctw** of console **240c** for exposing a top portion of bin **250**, and console openings **240coa**, **240cob**, and **240coc** may be provided through different portions of a side wall **240csw** of console **240c** for providing access to other portions of bin **250** and/or space **241s**. Bin **250** may be coupled (e.g., permanently or removably (e.g., such that an end user may replace, etc.)) to vehicle **290** (e.g., within console space **241s**) using any suitable coupling features **251** (e.g., snap features, clip features, track features, etc.).

[0080] FIGS. 3 and 3A-3N show an illustrative adjustable storage bin system **350** that may be the same as or similar to adjustable storage bin system **250** (e.g., bin **250a** or otherwise (e.g., any suitable bin **250b-250m**, etc.)). As shown, a carrier traveling assembly or carrier traveling tray or carrier traveling mat **358** (e.g., a thermoplastic and/or metal) of any suitable shape that may be

provided to selectively travel with respect to (e.g., relative to) a storage bin carrier **353**, such as within a carrier space **353s** of carrier **353** (e.g., carrier space **353s** may be any suitable three-dimensional shape as may be defined by the interior of any suitable carrier walls (e.g., top wall **353a**, bottom wall **353m**, left wall **353l**, right wall **353r**, front wall **353f**, and rear wall **353e** of carrier **353** (e.g., a thermoplastic and/or metal)) of carrier **353**), while adjustable storage bin system **350** may also include, in some embodiments, any suitable top surface component **352** (e.g., a soft top mat tray surface (e.g., textile and/or elastomer)) that may be provided along the top surface of tray or mat **358**. For example, mat **358** may be configured to travel (e.g., selectively travel) with respect to (e.g., relative to) storage bin carrier **353** amongst any suitable tray positions, such as a fully up position **358'** of FIG. 2K, a fully down position **358''** of FIG. 2K, an intermediate up position **358'''** of FIG. 2K, an intermediate down position **358''''** of FIG. 2K, and/or the like, using any suitable tray movement assembly (e.g., locking multi-step assembly **330**, push-push latch assembly **335**, storage bin motor assembly **390**, scissor-type assembly **581**, and/or any other suitable tray movement assembly that may be operative to control movement of the tray with respect to the carrier (e.g., to move the tray between and/or hold the tray at different tray positions with respect to the carrier)). Storage bin carrier **353** may be removably or permanently coupled to any suitable portion of any suitable vehicle (e.g., to center console **240c** of vehicle **290** (e.g., within any suitable console space **241s** (e.g., within bin space portion **241sb** of console space **241s**) that may be defined at least partially by console **240c** and/or any other suitable portion or vehicle bin component of vehicle **290**)). In some embodiments, storage bin carrier **353** may be formed as a portion of console **240c** (e.g., formed into console **240c** when the console is being formed (e.g., storage bin carrier **353** may be formed as a portion of a main carrier or storage bin of console **240c** (e.g., by injection molding, gluing, heat stake, etc.))) or may be mounted to console **240c** in any suitable way (e.g., bin carrier **353** may be mounted with positive attachments (e.g., fasteners, clips, screws, etc.) to any suitable portion of console **240c** (e.g., using any suitable coupling features **251**)). Spring compartments **354** and **356** may be provided along opposing portions (e.g., sides (e.g., left carrier side wall **353l** and right carrier side wall **353r**)) of storage bin carrier **353** (e.g., an open top box-like structure) for housing respective spring components **362** and **362a** (e.g., steel), each of which may be held between a bottom interior surface **353b** (e.g., of bottom carrier wall **353m**) of storage bin carrier **353** and a respective spring guide housing **361** and **361a** of a bottom surface of carrier traveling mat **358** for enabling any suitable spring-loaded (e.g., manual push-push type) mechanism for positioning mat **358** at various positions with respect to carrier **353** (e.g., along the Z-axis (e.g., positions (e.g., positions **358'**-**358''**)) that may be defined by a push-push-latch **364** that may be configured to engage with a push-push-latch tab **365** (e.g., as may be extending from bottom interior surface **353b** of carrier **353**) (e.g., an assembly **335** (e.g., push-push latch assembly)) and/or by stopper surface(s) **375** (e.g., as may be tab surface(s) extending out from a side interior surface **353r** of carrier side wall **353r** of carrier **353** or notch surfaces extending into a side interior surface of carrier **353**) for engaging with a locking pin or lever **374** (e.g., an assembly **330** (e.g., a locking multi-step assembly)))). Additional bumper component(s) **359** and/or **359a** (e.g., elastomer) may be positioned between a top surface of mat **358** and a bottom surface stopper **360/360a** of carrier **353** as control features for promoting smooth travel of mat **358** with respect to carrier **353**. Additionally or alternatively (e.g., instead of a manual mechanism for a user to physically manipulate mat **358** up or down with respect to carrier **353**), dampener track rails **355** and **357** may be provided along opposing portions (e.g., sides) of storage bin carrier **353** for interacting with respective carrier traveling mat dampener housing components **366a** and **366** of dampeners **367a** and **367** (e.g., silicone and/or a hard thermoplastic) that may be provided along opposing external portions (e.g., sides) of mat **358** (e.g., components **366/366a** may be supporting or physically coupled to or integrated with mat **358**) for enabling any suitable electronically controlled servo/motor **392** and/or motor microcontroller **394** of a storage bin motor assembly **390** of bin system **350** to physically manipulate dampeners **367/367a** up or down with respect to rails

355/357 and, thus, to physically manipulate mat 358 up or down with respect to carrier 353. For example, in some embodiments, a force may be applied directly on top mat surface 352 (e.g., a downward force in the direction of $-Z$ (e.g., manually by a user)) to adjust the position (e.g., using latch 364 (e.g., assembly 335) and/or a locking multi-step assembly (e.g., assembly 330)). Additionally or alternatively, the movement of mat 358 with respect to carrier 353 between any suitable positions may be controlled electronically through use of any suitable motor/servo 392 (e.g., in place of or in addition to rails 355/357 and/or dampeners 367a/367), which may be activated or otherwise controlled by a motor microcontroller 394 (e.g., automatically or by user manipulation of any suitable physical switch (e.g., switch 216b) or graphical user interface (e.g., touch screen 216a) of any suitable user interface (e.g., I/O interface 216 of subsystem 220f or otherwise of vehicle 290) that may be positioned anywhere within the vehicle) through any suitable system of the vehicle (e.g., via vehicle subsystem 220f or a mechanical switch located on a door trim or any other suitable user accessible location within the vehicle). If movement of the mat is configured to be movable via electronics (e.g., automatically and/or via user interface (e.g., rather than by a user's physical manipulation of the mat using a manual push-push type mechanism or otherwise), such as via storage bin motor assembly 390), then one or more sensors (e.g., sensors 215a/215b) of an electronic control system (e.g., control system or processor 212 (e.g., of subsystem 220f), which may include any suitable head unit 212a and/or any suitable body control unit 212b) may be positioned in any suitable position(s) with respect to the mat to determine if there are any obstacles blocking safe movement of the mat in one or more directions and utilize any suitable sensor data to automatically prevent certain electronic movement of the mat until the obstacle has been removed (e.g., a sensor 215b may be positioned below mat 358 (e.g., along surface 353b) or otherwise to determine if there are any distinct objects between sensor 215b and mat 358 (e.g., a stored object 5b of FIG. 2K) that may be an obstacle to downward (e.g., $-Z$) movement of the mat, and/or a sensor 215a may be positioned above mat 358 (e.g., along or adjacent surface 352) or otherwise to determine if there are any distinct objects between sensor 215a and mat 358 (e.g., a stored object 5a of FIG. 2K) that may be an obstacle to upward (e.g., $+Z$) movement of the mat). Such sensor(s) may be any suitable sensors, including, but not limited to, a weight sensor, a light sensor, a motion sensor, a camera, and/or the like. Therefore, adjustable storage bin system 350 may utilize a manual push-push, spring loaded type mechanism to maintain and push a tray (e.g., mat 358) into position (e.g., with additional bumpers and/or dampeners as control features for travel smooth up and down) or an electronic controlled servo/motor with the addition of guides to adjust the height of the tray with respect to a storage bin carrier 353. Any suitable power source (e.g., a power source 217 of subsystem 220f) may be configured to power control system 212 and/or user interface 216 and/or motor assembly 390 and/or sensors 215a/215b (see, e.g., FIG. 1D).

[0081] FIG. 3A shows spring components 362 and 362a, locator spring housings 361 and 361a, softer rubber stopper elements 359 and 359a, and a primary surface (e.g., an interior bottom surface 353a is of top wall 353a that may be opposite interior top surface 353b of bottom wall 353m) of carrier 353 providing or coupled to surface stoppers 360 and 360a. Moreover, any suitable device modules 363 and 363a of system 350 may be illustrated in FIG. 3A and may be used to functionally couple to (e.g., via mat 358 (e.g., via top mat surface 352)) respective portable user electronic devices 220u and 220ua (e.g., to electrically couple thereto (e.g., for charging power component(s) of the device(s) (e.g., using any suitable power supply of the vehicle)) and/or to communicatively couple thereto (e.g., for communicating any suitable data between devices (e.g., for communicatively coupling device 220u and/or device 220ua with any other suitable device(s) (e.g., vehicle subsystem 220f), such as for synchronizing a portable user device with a vehicle infotainment system device))). Module 363 and/or module 363a may be any suitable wireless charging module and/or physical connector and may be coupled (e.g., electrically (e.g., via any bus 18 or otherwise) to any suitable electronic subsystem of vehicle 290 (e.g., subsystem 220f)). When

a portable user electronic device (e.g., device **220u** and/or device **220ua** (e.g., a smartphone, tablet, etc.)) is positioned on mat **358** of adjustable storage bin system **350**, the user device may be electrically coupled to any other suitable subsystem of vehicle **290** (e.g., via module **363** and/or module **363a**). For example, one or more suitable data and/or power connectors (e.g., USB-C connector) may be provided by module **363/363a** of adjustable storage bin system **350** for interacting with the user device(s). In some embodiments, module **363/363a** may be an I/O component **16** of any suitable vehicle subsystem (e.g., vehicle subsystem **220f**) that may be any suitable data and/or power connector (e.g., a Universal Serial Bus (“USB”) connector or any other suitable connector type (e.g., male or female)) that may be physically connected or otherwise coupled to a connector (e.g., female or male) of user device **220u** when user device **220u** is positioned on mat **358** in a mating position with such a connector of adjustable storage bin system **350** and/or an I/O component **16** of any suitable vehicle subsystem (e.g., vehicle subsystem **220f**) that may be any suitable data and/or power connector (e.g., a wireless charger (e.g., an inductive charging pad or the like)) that may be functionally coupled to user device **220u** when user device **220u** is positioned on mat **358** of adjustable storage bin system **350**. Any suitable bus(es) **18** may communicatively/functionally couple device modules **363** and **363a** to any suitable other components of any suitable vehicle subsystem (e.g., processor **12**, communications component **14**, and/or power supply **17** of vehicle subsystem **220f**). Therefore, when positioned on mat **358** of adjustable storage bin system **350**, not only may user device **220u** be supported (e.g., physically) by adjustable storage bin system **350**, but device **220u** may also be coupled (e.g., electrically, communicatively, power chargingly, or otherwise functionally) to any vehicle subsystem(s) via any suitable I/O component(s) within or adjacent storage bin system **350** for any suitable purpose(s) (e.g., charging of device **220u** and/or synchronization (e.g., media synchronization) of user device **220u** with any vehicle subsystem (e.g., synchronization of a screen of vehicle subsystem **220f** with a screen of user device **220u** or otherwise)). In some embodiments, once a user device **220u** is functionally positioned on mat **358** (e.g., once device **220u** begins charging with module **363** and is communicatively coupled to (e.g., synchronized with) a vehicle subsystem (e.g., subsystem **220f**, such that the device may be functionally used via subsystem **220f** (e.g., such that a user interface of user device **220u** may be transferred to a user interface of vehicle subsystem **220f** such that any suitable data and/or functionality of user device **220u** may be utilized by vehicle subsystem **220f** and/or such that any suitable power and/or data of vehicle subsystem **220f** may be transferred to user device **220u**))) or otherwise detected to be properly positioned on mat **358** for any suitable amount of time (e.g., using any suitable sensor(s) of module(s) **363/363a** and/or the like (e.g., sensor(s) **215a/215b**)), the system may be configured to automatically lower mat **358** down into the carrier to initiate any suitable deep storage mode (e.g., the system may automatically detect the functional positioning of a phone on the mat and then automatically lower the mat so that the phone is moved out of view of the user and more carrier storage space is made available above the mat).

[0082] FIG. 3B shows dampener track rails **355** and **357**, respective contact surfaces identified as track rail guides **368** and **368a**, and bin carrier traveling mat dampener housings **366** and **366a** that may be used to secure respective dampeners **367** and **367a**, which may travel through dampener track rails **355** and **357**, respectively. Additionally, a push-push-latch holder **364h** with push push-latch **364** may be provided at a bottom surface of mat **358** while push-push-latch-tab **365** may be provided at an interior bottom surface **353b** of carrier **353** or otherwise to provide a push-push latch assembly **335** that may be configured to secure mat **358** with respect to carrier **353** in a bottom position. Push push-latch **364** may be configured to hold mat **358** in a particular position with respect to carrier **353** (e.g., any suitable position(s), such as fully down position **358''** of FIG. 2K (e.g., a fully down storage position), intermediate up position **358''** of FIG. 2K, intermediate down position **358''** of FIG. 2K, and/or the like) while engaging push-push-latch-tab **365**, then it may be pushed again (e.g., by a user via mat **358** (e.g., in the -Z direction)) passed the fully down position **358''** to a bottom release position such that tab **365** may release (e.g., disengage) latch **364** whereby

mat 358 may be released back up (e.g., in the +Z direction) to fully up position 358'. [0083] FIGS. 3C-3F''' show a front edge of adjustable storage bin device 350, revealing a locking multi-step assembly 330 of device 350 that may be configured to enable mat 358 to stop at different levels or positions (e.g., fully up position 358' of FIG. 2K, intermediate up position 358'' of FIG. 2K, intermediate down position 358''' of FIG. 2K, and/or the like) before engaging push-push latch assembly 335 (e.g., before engaging together push-push-latch 364 and tab 365) for holding mat 358 at a fully down position (e.g., for holding mat 358 at fully down position 358'' of FIG. 2K (e.g., a fully down storage position)). A pivot axis in mat 358 (e.g., a carrier traveling lever axis feature 370 of mat 358) may be rotatably holding a lever guide 372 (e.g., a thermoplastic and/or metal) and a spring loaded component 371 (e.g., a spring (e.g., steel) that may bias guide 372 for a particular rotation about axis feature 370 of mat 358 (e.g., when feature 370 may extend through a guide opening 3720 in guide 372)), whereby guide 372 may rotate about feature 370 (e.g., about an X-axis), while a lever spring 373 may be positioned within a slot 372s in guide 372 to bias a lever 374 at least partially in slot 372s (e.g., outward (e.g., in -X direction)) to secure a position of mat 358 with respect to one of potentially multiple different stopper surface(s) 375 (e.g., steel). As shown in a progression of assembly 330 of device 350 from a configuration of FIGS. 3D and 3D' (e.g., fully up position 358') to a next configuration of FIGS. 3E and 3E' to a next configuration of FIGS. 3F and 3F' (e.g., intermediate up position 358'') as mat 358 is moved downward (e.g., in the -Z direction) with respect to carrier 353, guide 372 may rotate (e.g., as a cam) about feature 370 of mat 358 in a first direction (e.g., direction CCW) (e.g., against a bias of component 371) when lever 374 may interact with an angled surface 375aa of a stopper 375 as mat 358 is moved downward with respect to carrier 353 and stopper 375 (e.g., against a bias of spring component(s) 362/362a), and then guide 372 may rotate back in a second direction (e.g., direction CW) opposite the first direction (e.g., using a bias of component 371) when lever 374 has moved down beyond stopper 375, such that a bottom (e.g., flat) surface 375af of stopper 375 may interact with lever 374 to hold mat 358 in a position with respect to stopper 375, even when a downward force on mat 358 may be terminated while lever 374 is biased towards and against side interior surface 353r is under stopper 375 (e.g., by spring 373 and spring component(s) 362/362a). However, when lever 374 is pulled away from side interior surface 353r is and from under stopper 375 (e.g., in a +X direction further into slot 372s (e.g., using any suitable user controlled mechanism)), lever 374 may disengage with stopper 375 and mat 358 may be able to move upward (e.g., in the +Z direction (e.g., due to bias of spring component(s) 362/362a)) without interference from stopper 375. Alternatively, as shown in FIGS. 3F'' and 3F''', system 350 may be configured such that lever 374 may be pushed further into side interior surface 353r is (e.g., in a -X direction (e.g., using the bias of spring 373)) when mat 358 is pushed further downward (e.g., in the -Z direction opposite the bias of spring component(s) 362/362a (e.g., by a user's manual downward pressure)) beyond the bottom surface of the lowest stopper (e.g., lower stopper 375b), whereby lever 374 may be aligned with and be pushed into a deep return track 375t that may be formed into side interior surface 353r is in the -X direction beyond the depth of a stopper track 375s in side interior surface 353r is adjacent stoppers 375 (e.g., when mat 358 is pushed further downward (e.g., in the -Z direction opposite the bias of spring component(s) 362/362a) beyond the bottom surface of the lowest stopper (e.g., lower stopper 375b), the free end of lever 374 may transition from abutting stopper track 375s of side interior surface 353r is in the -X direction to abutting the deeper return track 375t of side interior surface 353r is in the -X direction). Then, once a downward pressure on mat 358 is released (e.g., by a user), the upward bias of spring component(s) 362/362a may push mat 358 upwards (e.g., in the +Z direction) whereby lever 374 may travel upwards along return track 375t rather than along stopper track 375s, such that lever 374 may not be stopped by stopper(s) 375. A connector track 375c in side interior surface 353r is may enable lever 374 to transition from the depth of the top of track 375t to the depth of the top of track 375s for returning mat 358 to its top position 358'. There may be any suitable number of different

stoppers **375** (e.g., spaced apart along the Z-axis) for defining different positions of mat **358** with respect to carrier **353** (e.g., an intermediate up position **358"** (e.g., with an upper stopper **375a**), an intermediate down position **358"** (e.g., with a lower stopper **375b**), and/or the like). Dampener housing components **366a** and **366** of dampeners **367a** and **367** may be configured as dampeners to control speed of a system for moving mat **358** with respect to carrier **353** (e.g., along the Z-axis) and/or to control tracking (e.g., a gear may be provided to run on a silicon component to ensure the mat moves up and down rather than side to side, where a tactile softness may be enabled). In some embodiments, a dampener may be substituted by a powered motor in an electric assembly, which may not need assembly **330** and/or assembly **335**. In some embodiments, both an electrical movement assembly (e.g., assembly **390**) and a mechanical movement assembly (e.g., assembly **330**, **335**, etc.) may be provided in system **350** and either may be selectively used by a user at any given time.

[0084] While mechanical locking multi-step assembly **330** may only be shown along one side of carrier **353** (e.g., stopper(s) **375** along the -X side wall **353r**), it is to be understood that another mechanical locking multi-step assembly may additionally or alternatively be provided on the opposite side (e.g., along the +X side wall **353l**) and/or any additional area(s) (e.g., Y-side wall(s), corner(s), center, etc.) to provide a differently positioned and/or more robust solution. While spring compartments **354** and **356** and respective springs **362a** and **362** of a spring assembly **369** may only be shown along opposing portions (e.g., sides) of carrier **353** (e.g., opposing sides along the X-axis (e.g., side walls **353l** and **353r** of carrier **353**)) for providing Z-axis spring bias to mat **358** with respect to carrier **353**, it is to be understood that such spring(s) may be provided in any additional or alternative locations with respect to carrier **353**, such as opposing sides along the Y-axis, in each of the four corners, in the center (e.g., about latch **364** and tab **365**), and/or the like. While latch **364** and tab **365** may only be shown extending along the Z-axis within the center of carrier **353** for providing latch assembly **335**, it is to be understood that such latch assembly components may be provided in any additional or alternative locations with respect to carrier **353**, such as along one side of the carrier, in a corner of the carrier, or two or more such latch assemblies may be provided for working in tandem to provide a more robust solution (e.g., in two opposite corners or along two opposing walls, etc.). While dampener housing components **366a/366** of dampeners **367a/367** working with rails **355/357** and guides **368a/368** may only be shown along opposing portions (e.g., sides) of carrier **353** (e.g., opposing sides along the Y-axis (e.g., side walls **353e** and **353f** of carrier **353**)), it is to be understood that such dampener housing component(s), dampener(s), track(s), and rail(s) of a dampener assembly **351** may be provided in any additional or alternative locations with respect to carrier **353**, such as opposing sides along the X-axis, on each of the four sides, in the four corners, and/or the like.

[0085] In some embodiments, by positioning mechanical locking multi-step assembly **330**, springs of spring assembly **369**, dampener guides/rails of dampener assembly **351**, and/or latch assembly **335** in or near corners and/or in a center of carrier **353**, rather than along a middle portion or otherwise of one or more sides **353l**, **353r**, **353f**, and **353e** of carrier **353**, one or more side storage bin openings may be more easily provided through one or some or each of sides **353l**, **353r**, **353f**, and **353e** of carrier **353** for enabling the passage of any suitable objects (e.g., user device **220u**, an object **5a**, an object **5b**, etc.) therethrough. In other embodiments, a side storage bin opening may be provided through a carrier side wall along which dampener guides/rails of a dampener assembly may also be provided and/or a side storage bin opening may be provided through a carrier side wall along which stopper surfaces of a locking multi-step assembly may also be provided.

[0086] FIGS. **4** and **4'-4"** show another illustrative adjustable storage bin system **450** that may be the same as or similar to adjustable storage bin system **250** (e.g., bin **250a** or otherwise (e.g., any suitable bin **250b-250m**, etc.)) and/or storage bin system **350**. As shown, adjustable storage bin system **450** may include a carrier traveling mat **458** that may be provided to selectively travel with respect to a storage bin carrier **453**, which may be removably or permanently coupled to any

suitable portion of any suitable vehicle (e.g., center console **240c** of vehicle **290**). Bin system **450** may include dampener guides/rails of a dampener assembly **451** on opposing sides **453e** and **453f** of carrier **453**, spring compartments and respective springs of a spring assembly **469** on opposing sides **4531** and **453r** of carrier **453**. However, unlike bin system **350**, bin system **450** may be shown to include at least one side storage bin opening provided through a side wall of carrier **453**. For example, as shown in FIGS. **4** and **4''**, a side storage bin opening **481** may be provided through (e.g., extends through a portion of) side wall **453e** of carrier **453**, which may include any suitable dampener guides/rails **455/468a** of dampener assembly **451**, while a side storage bin door **476** (e.g., a thermoplastic and/or metal) may be provided to selectively reside within opening **481** for defining any suitable functional features **483** (e.g., a thermoplastic and/or metal) that may functionally replace any suitable portions of dampener guides/rails of dampener assembly **451** that might otherwise be defined by the portion of side wall **453e** taken up by opening **481**, whereby features **483** of door **476** and guides/rails **455/468a** of wall **453e** may together seamlessly provide the functionality for dampener assembly **451** when door **476** is in its closed configuration within opening **481** (alternatively, in some embodiments, features (e.g., features **483**) of a door and spring compartment(s) or otherwise of any suitable spring assembly of any suitable wall (e.g., spring assembly **469** of wall **4531**) and/or stopper surface(s) (e.g., surface(s) **375**) of any suitable wall may together seamlessly provide the functionality for a push-push latch and/or locking multi-step assembly when the door is in its closed configuration within a door opening). Although opening **481** may be shown to span a majority of the X-axis width of wall **453e** (e.g., intersecting dampener assembly **451**), opening **481** may alternatively span a portion of the X-axis width of wall **453e** that does not intersect dampener assembly **451** (e.g., a width portion between assembly **451** and a Z-axis edge of wall **453e**), such that door **476** may not include any functional features **483** as it may not intersect or interfere with assembly **451** of bin system **450**. While side storage bin opening **481** may only be shown to be provided through side wall **453e** of carrier **453** of bin system **450** (e.g., a side wall in a X-Z plane), it is to be understood that a side storage bin opening may be additionally or alternatively provided through any other suitable portion of a carrier of a bin system, such as through a perpendicular side wall (e.g., a side wall in a Y-Z plane, as may be shown by opening **481'** through wall **4531** of bin system **450'** of FIGS. **4A** and **4A'-4A''**) and/or through two or more side walls of the same bin (e.g., a first opening **481** through side wall **453e** and a second opening through side wall **4531** of the same bin).

[0087] Door **476** may be selectively opened or closed with respect to opening **481** for enabling any suitable objects to be passed through opening **481** between a carrier space (e.g., as may be defined by the interior surfaces of carrier walls **453a**, **453m**, **453r**, **4531**, **453f**, **45e**, and/or the like) and an environment external to carrier **453** when door **476** is in an open position. Any suitable door movement assembly **489** may be provided for enabling door **476** to move between its open and closed positions with respect to opening **481**. For example, carrier **453** may include door axis tabs **478** and **478a** (e.g., a thermoplastic and/or metal) of an assembly **489** that may be positioned on any two opposing sides of opening **481** (e.g., extending outwardly from carrier **453** about opposing upper external sides of opening **481**) and that may support a door axis feature **477** (e.g., steel) of assembly **489** extending along a rotation axis R between tabs **478** and **478a**, whereby feature **477** may be supported by yet free to rotate (e.g., about an X-axis) with respect to tabs **478** and **478a**, and feature **477** may be integrated into or coupled to door **476** such that door **476** may rotate with feature **477** about rotation axis R for opening or closing door **476** with respect to opening **481**. Feature **477** may be spring biased to rotate for opening door **476** outwardly away from carrier **453** but may be held closed by any suitable push-push latch/tab or otherwise. Any other suitable type(s) of door movement assembly may be utilized for selectively opening and/or closing door **476** with respect to opening **481** (e.g., mechanical and/or electrical mechanism(s) (e.g., servo/motor) that may allow a user to selectively open or close the door via physical force or user interface engagement and/or that may allow the door to be manipulated automatically (e.g., in response to

certain events (e.g., when mat **458** lowered to a position aligning with opening **481**, a cam may be configured to release a spring that may push open door **476**)). As another example, assembly **489** may be configured to provide a counter force to bias door **476** in a closed position with respect to opening **481** (e.g., with a spring), such that functional features **483** may be used by dampener assembly **451** when mat **458** may be traveling up and down within carrier **453** (e.g., along the Z-axis) and/or such that a user's fingers may not be dangerously passed through opening **481** when the mat is being moved, whereby the mat may travel up and down with dampened velocity, and such that door **476** may be opened (e.g., rotated about axis R) to allow objects to pass through opening **481** from mat **458** or otherwise within the carrier space to an environment external to the carrier (e.g., to an alternative storage zone (e.g., remaining portion **241sr** of console space **241s**) and/or to a space external from the console (e.g., to remove the object from the vehicle altogether) when the mat is not moving).

[0088] As shown in FIG. 2K, bin system **450** may be used to position mat **458** (e.g., up or down along mat movement axis MMA (e.g., along a Z-axis)) to a position aligned with opening **481** (e.g., to position **358'''**) such that the mat and/or any device(s) thereon may then be moved out from carrier space **353s** into remaining portion **241sr** of console space **241s** via opening **481** (e.g., sideways along mat movement axis MSA (e.g., along a Y-axis)), and then the object may be removed from portion **241sr** of console space (e.g., into ambient space **298as** external from console space **241s** (e.g., a portion of space **290us**)) via console opening **240coa** (e.g., along an X-axis). Console opening **240coa** may be protected by its own door **240coad** that may be hidden or lockable to further hide or protect portion **241sr** of console space **241s** until it is to be accessed via opening **240coa** by a user (e.g., door **240coad** may be controlled by mechanically and/or electronically by a user and/or automatically due to any suitable events). The mat or any other objects (e.g., device **220ua** on the mat or an object **5b** below the mat) may be removed from the carrier into space **241sr** (e.g., onto support surface **241srl**) manually by a user via opening **481** and either opening **240coa** or opening **240com** or automatically via opening **481** by any suitable mat/object ejection assembly **299** (e.g., an assembly (e.g., electromechanical mechanism) configured to eject the mat outwardly along axis MSA automatically or in response to some user interface event). For example, a user could extend their arm down through opening **240com** and slide device **220ua** off of the mat in position **358'''** in a-Y direction along axis MSA through opening **481** and onto support surface **241srl**, then the mat could be raised to position **358'** such that the device may only be accessible on support surface **241srl** via opening **240coa** or only after the mat was lowered back to position **358'** or lower. This may provide some additional security for the device. In such embodiments, that bin opening may not be selectively closed by a door, but instead the bin opening may remain open and console opening **240coa** may be provided with a door that may securely lock or otherwise hide the bin opening and support surface **241srl** when not being accessed by the user. Additionally or alternatively, as shown in FIGS. 2K and 4A-4A'', an opening **481'** may be provided via wall **4531** of bin **450'**, such that an object may be removed from the mat at position **358''** via that bin opening (e.g., along the X-axis) and then via console opening **240cob** (e.g., also along the X-axis). In such embodiments, that bin opening may not be selectively closed by a bin door, but instead the bin opening may remain open and console opening **240cob** may be provided with a door **240cobd** that may securely lock or otherwise close to hide the bin opening when not being accessed by the user via opening **240cob** (e.g., door **240cobd** may be controlled by mechanically and/or electronically by a user and/or automatically due to any suitable events).

[0089] FIGS. 5 and 5A-5C show another illustrative adjustable storage bin system **550** that may be the same as or similar to adjustable storage bin system **250** (e.g., bin **250a** or otherwise (e.g., any suitable bin **250b-250m**, etc.)) and/or storage bin system **350** and/or storage bin **450** and/or storage bin **450'**. Adjustable storage bin system **550** may include a scissor-type assembly **581** to achieve adjustability of a mat **558** (e.g., with or without module(s) **563/563a**) with respect to a carrier **553** (e.g., up and down along the Z-axis). Scissor-type assembly **581** may include a scissor slide plate

upper component **582** (e.g., a thermoplastic and/or metal), a scissor slide plate lower component **583** (e.g., a thermoplastic and/or metal), a strut component **584** (e.g., a gas loaded (e.g., gas pressurized) or fluid loaded strut (e.g., metal) component), a scissor sliding plate **585**, and a push-push latch **564**. A pin **584p1** at a first end of strut **584** may be rotatably positioned within a slot **585a** of scissor sliding plate **585** (e.g., rotatable about an X-axis), a pin **584p2** at a second end of strut **584** (e.g., at a free end of a piston **584t** of strut **584**) may be rotatably positioned within a slot **582b** positioned along a length of scissor slide plate upper component **582** (e.g., rotatable about an X-axis), a pin **582p2** at a first end of scissor slide plate upper component **582** may be rotatably positioned within a slot **558a** of mat **558** (e.g., rotatable about an X-axis), a pin **582p1** at a second end of scissor slide plate upper component **582** may be slidably positioned within a slide slot **585s** of scissor sliding plate **585** (e.g., slidable along a Y-axis), a pin **583p1** at a first end of scissor slide plate lower component **583** may be rotatably positioned within a slot **585b** of scissor sliding plate **585** (e.g., rotatable about an X-axis), a pin **583p2** positioned along a length of scissor slide plate lower component **583** may be rotatably positioned within a slot **582a** positioned along a length of scissor slide plate upper component **582** (e.g., rotatable about an X-axis), and a pin **583p3** at a second end of scissor slide plate lower component **583** may be slidably positioned within a slide slot **558s** of mat **558** (e.g., slidable along a Y-axis). Assembly **581** may be configured to adjust its height, such as from height H of FIG. 5B to height H' of FIG. 5C (e.g., for lowering the position of mat **558** in the -Z direction within carrier **553**). When no load is applied on mat **558** (e.g., when no load P is applied downwardly on a top surface component **552** of mat **558** in the -Z direction), strut **584** may be configured to extend a piston **584t** (e.g., in a direction U) to achieve a fully up position of assembly **581** (e.g., "full up"). When a load P is applied downwardly on mat **558** (e.g., on top surface component **552** of mat **558** in the -Z direction (e.g., by a user)), the force may be transmitted through mat **558** and assembly **581**, whereby pin **583p3** of scissor slide plate lower component **583** may slide in the +Y direction along slot **558s** of mat **558** and whereby pin **582p1** of scissor slide plate upper component **582** may slide in the +Y direction along slot **585s** of scissor sliding plate **585**, which may compress piston **584t** (e.g., in a direction D) into strut **584** and reduce the height of the assembly (e.g., from height H of FIG. 5B to reduced height H' of FIG. 5C to achieve a down position of assembly **581**). Push-push latch **564** may engage with a push-push tab to hold assembly **581** in the down position until later disengagement and removal of any load P, such that piston **584t** may extend (e.g., in direction U) to move assembly **581** back to its full up position. A benefit of assembly **581** may be that it relies on geometry of axes, no dampeners, less springs, and/or less complexity than other mat moving assemblies. Carrier **553** may be provided with openings through side walls without impacting the functionality of assembly **581**.

[0090] Any suitable types of objects (e.g., purses, glasses, food, etc.) may be stored in various locations with respect to an adjustable storage bin system. As shown in FIG. 2K, any suitable object **5c** may be stored within console space **241s** in a position below or otherwise outside of the bin carrier (e.g., in space **241sr** (e.g., on or below surface **241srl**)) as may be accessible by a user via console opening **240coc** or console opening **240coa** (e.g., console opening **240coc** may be provided with a door **240cocd** that may securely lock or otherwise close to hide the opening when not being accessed by the user (e.g., door **240cocd** may be controlled by mechanically and/or electronically by a user and/or automatically due to any suitable events)), any suitable object **5b** may be stored within the carrier space of the bin (e.g., within space **241s**) below the mat as may be accessible by a user via console opening **240cob** or via console opening **240coa** and a bin opening (e.g., opening **481**), any suitable object **220ua** or otherwise may be stored on a top surface of a mat (e.g., on surface **352**) as may be accessible by a user via console opening **240com** regardless of the mat's position within the carrier space and/or via console opening **240cob** and a bin opening (e.g., opening **481**) when the mat is positioned below or at that bin opening and/or via console opening **240coa** and a bin opening (e.g., opening **481**) when the mat is positioned below or at that bin opening, and/or any suitable object **5a** may be stored above the mat when the mat is lowered at

least a certain amount below its fully up position (e.g., within a portion of the carrier space above the mat using any suitable holding apparatus **352'** (e.g., a cargo net or retractable sliding support surface) that may be suspended (e.g., automatically or by the user) above the mat (e.g., using hooks along interior side walls of the carrier) when the mat is positioned below the suspension points **352s'** for the holding apparatus and that may be retracted or removed (e.g., automatically or by the user) when the mat is to be accessed or moved upwards)) as may be accessible by a user via console opening **240com**. Carrier traveling mat **358** and module(s) **363** may be removably coupled to the remainder of adjustable storage bin system **350** (e.g., to carrier **353**) such that mat **358** and module(s) **363** may be passed through any suitable bin opening (e.g., opening **481**, **481'**, etc.). For example, once moved to position **358''**, mat **358** and module(s) **363** may be passed from carrier space **353s** through opening **481** into console space **241sr** (e.g., onto support surface **241srl**) manually by a user via either opening **240coa** or opening **240com** or automatically by any suitable mat ejection assembly (e.g., an assembly (e.g., electromechanical mechanism) configured to eject the mat outwardly along axis MSA automatically or in response to some user interface event). In some embodiments, module(s) **363** may remain electrically coupled to any suitable data and/or power sources for maintaining a functional relationship with any user device(s) **220ua** that may be positioned on the mat when the mat is at least partially removed from carrier space **353s** (e.g., the mat and module and user device may remain on support surface **241srl** and continue charging a user's device while other items may be stored within carrier space **353s**). In other embodiments, only object(s) (e.g., user device **220u**) may be ejected through the carrier opening while the mat may remain within the carrier space.

[0091] In a particular use case, if a user is not positioning a user device **220ua** on the mat, the mat may be lowered to its fully down position to maximize the amount of carrier space **353s** accessible to a user above the mat. In some embodiments, a door movement assembly (e.g., assembly **489**) may be configured like a garage door opener, whereby door **476** may translate from a closed position in an X-Z plane within opening **481** to an open position in an X-Y plane extending across space **353s** within carrier **353** (e.g., as shown by open door **4760** of FIG. 2K), such that the open door may provide a shelf surface on which a user may position any suitable objects for storage thereon (e.g., the system may be configured to automatically open the door or otherwise provide such a shelf above the mat once the mat is lowered to an intermediate or fully down position (e.g., to protect any device(s) on the mat and to provide a sturdy surface for storing more objects)). In some embodiments, a carrier may have a side storage bin opening without an associated door, whereby the carrier space may be accessed by a user via that a side storage bin opening and a proximate console opening, which may or may not have its own associated door (e.g., opening **240cob** may not include a door **240cobd** and opening **481'** may not include a door **476'**, such that openings **240cob** and **481'** may always be exposed to one another and the carrier space may always be accessible via openings **240cob** and **481'** by a user).

[0092] In some embodiments, there is provided a vehicle (e.g., vehicle **290**) that may include a vehicle bin component (e.g., console **240c**) defining a vehicle bin component space (e.g., space **241a**), and a storage bin system (e.g., system **250**) that may include a bin carrier (e.g., carrier **353**) defining a carrier space (e.g., space **353s**), a tray (e.g., tray **358**), and a tray movement assembly (e.g., assembly **330**, assembly **335**, assembly **390**, assembly **581**, and/or the like) operative to control movement within the carrier space of the tray between a plurality of tray positions relative to the bin carrier, wherein the bin carrier may be at least partially positioned within the vehicle bin component space. In some embodiments, the vehicle may further include an electronic vehicle subsystem (e.g., subsystem **220f**), the storage bin system may further include a device module (e.g., module **363**), the device module may be physically coupled to the tray, the device module may be electrically coupled to the electronic vehicle subsystem, and the device module may be operative to electrically couple to a user electronic device (e.g., device **220u**) when the user electronic device is positioned on the tray. In some embodiments, the device module may include a wireless charging

module. In some embodiments, the device module may include a USB connector. In some embodiments, the vehicle bin component may include a center console (e.g., console **240c**). In some embodiments, the vehicle bin component may include one of the following: a glove box (e.g., box **249g**); an interior storage bin (e.g., box **249g**); a door trim (e.g., door trim portion **249h**); an overhead storage bin (e.g., portion **249i**); a trunk trim quarter garnish (e.g., garnish **249j**); a floor panel (e.g., portion **249k**); a seat (e.g., portion **249l**); an upper panel (e.g., panel **249m**); or an instrument panel storage (e.g., panel **249m**). In some embodiments, the vehicle may be one of the following: a road vehicle; a rail vehicle; a watercraft; or an aircraft. In some embodiments, the tray movement assembly comprises a push-push latch assembly (e.g., assembly **335**). In some embodiments, the tray movement assembly comprises a locking multi-step assembly (e.g., assembly **330**). In some embodiments, the tray movement assembly comprises a scissor-type assembly (e.g., assembly **581**). In some embodiments, the tray movement assembly comprises a storage bin motor assembly (e.g., assembly **390**). In some embodiments, the vehicle bin component space may include a bin space portion (e.g., portion **241sb**) and a remaining space portion (e.g., portion **241sr**), the bin carrier may be at least partially positioned within the bin space portion, the bin carrier may include a plurality of carrier walls (e.g., top wall **353a**, bottom wall **353m**, left wall **353l**, right wall **353r**, front wall **353f**, and rear wall **353e** of carrier **353**) that may at least partially define the carrier space, a side storage bin opening (e.g., opening **481**) that may extend through a carrier wall of the plurality of carrier walls, and the side storage bin opening may couple the carrier space with the remaining space portion. In some embodiments, a tray position (e.g., position **358**) of the plurality of tray positions may be aligned with the side storage bin opening. In some embodiments, the storage bin system may further include an ejection assembly (e.g., assembly **299**) that may be configured to eject at least one of the following from the carrier space and into the remaining space portion via the side storage bin opening once the tray is at the tray position: the tray; or an object (e.g., device **220ua**) positioned on the tray. In some embodiments, the vehicle bin component space may include a bin space portion and a remaining space portion, the bin carrier may be at least partially positioned within the bin space portion, the vehicle bin component may include a plurality of vehicle bin component walls that may at least partially define the vehicle bin component space, a vehicle bin component opening (e.g., opening **240coa**) that may extend through a vehicle bin component wall (e.g., wall **240csw**) of the plurality of vehicle bin component walls, and the vehicle bin component opening may couple the remaining space portion with an ambient space (e.g., space **298as**) that is external to the vehicle bin component space. In some embodiments, the vehicle may also include a door (e.g., door **240coad**), and a door movement assembly operative to control movement of the door between a first door position within the vehicle bin component opening and a second door position out from the vehicle bin component opening. In some embodiments, the storage bin system may further include a door (e.g., door **476**), and a door movement assembly (e.g., assembly **489**) that may be operative to control movement of the door between a first door position within the side storage bin opening and a second door position out from the side storage bin opening. In some embodiments, when the door (e.g., door **476**) is in the second door position, a portion of the door (e.g., portion **4760**) is positioned within the carrier space and is operative to support a user object (e.g., object **5a**).

[0093] In some embodiments, there is provided a vehicle (e.g., vehicle **290**) that may include a vehicle bin component (e.g., console **240c**) defining a vehicle bin component space (e.g., space **241a**), and a storage bin system (e.g., system **250**) that may include a bin carrier (e.g., carrier **353**) defining a carrier space (e.g., space **353s**), a tray (e.g., tray **358**), and a tray movement assembly (e.g., assembly **330**, assembly **335**, assembly **390**, assembly **581**, and/or the like) that may be operative to move the tray within the carrier space along a first axis (e.g., axis MMA) and move the tray out from the carrier space along a second axis (e.g., axis MSA) that is perpendicular to the first axis, wherein the bin carrier may be at least partially positioned within the vehicle bin component space.

[0094] In some embodiments, there is provided a vehicle (e.g., vehicle **290**) that may include a vehicle bin component (e.g., console **240c**) defining a vehicle bin component space (e.g., space **241a**), and a storage bin system (e.g., system **250**) that may include a bin carrier (e.g., carrier **353**) defining a carrier space (e.g., space **353s**), a tray (e.g., tray **358**), and a tray movement assembly (e.g., assembly **330**, assembly **335**, assembly **390**, assembly **581**, and/or the like) that may be operative to control movement within the carrier space along a first axis (e.g., axis MMA) of the tray between a first tray position (e.g., position **358'**) and a second tray position (e.g., position **358'** or position **358''**), wherein the bin carrier may include a wall (e.g., wall **3531**) with an opening (e.g., opening **481**) for enabling a user to move the tray along a second axis perpendicular (e.g., axis X) to the first axis from the second tray position to a third tray position external from the carrier space, wherein the bin carrier may be at least partially positioned within the vehicle bin component space.

[0095] One, some, or all of the processes described with respect to FIGS. **1-5C** may each be implemented by software, but may also be implemented in hardware, firmware, or any combination of software, hardware, and firmware. Instructions for performing these processes may also be embodied as machine- or computer-readable code recorded on a machine- or computer-readable medium. In some embodiments, the computer-readable medium may be a non-transitory computer-readable medium. Examples of such a non-transitory computer-readable medium include but are not limited to a read-only memory, a random-access memory, a flash memory, a CD-ROM, a DVD, a magnetic tape, a removable memory card, and a data storage device (e.g., memory **13** of a subsystem). In other embodiments, the computer-readable medium may be a transitory computer-readable medium. In such embodiments, the transitory computer-readable medium can be distributed over network-coupled computer systems so that the computer-readable code is stored and executed in a distributed fashion. For example, such a transitory computer-readable medium may be communicated from one subsystem to another directly or via any suitable network or bus or the like. Such a transitory computer-readable medium may embody computer-readable code, instructions, data structures, program modules, or other data in a modulated data signal, such as a carrier wave or other transport mechanism, and may include any information delivery media. A modulated data signal may be a signal that has one or more of its characteristics set or changed in such a manner as to encode information in the signal.

[0096] Any, each, or at least one module or component or subsystem of the disclosure may be provided as a software construct, firmware construct, one or more hardware components, or a combination thereof. For example, any, each, or at least one module or component or subsystem of system **1** may be described in the general context of computer-executable instructions, such as program modules, that may be executed by one or more computers or other devices. Generally, a program module may include one or more routines, programs, objects, components, and/or data structures that may perform one or more particular tasks or that may implement one or more particular abstract data types. The number, configuration, functionality, and interconnection of the modules and components and subsystems of system **1** are only illustrative, and that the number, configuration, functionality, and interconnection of existing modules, components, and/or subsystems may be modified or omitted, additional modules, components, and/or subsystems may be added, and the interconnection of certain modules, components, and/or subsystems may be altered.

[0097] The terminology used in the description of the various described embodiments herein is for the purpose of describing particular embodiments only and is not intended to be limiting. As used in the description of the various described embodiments and the appended claims, the singular forms “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. The term “and/or” as used herein may refer to and encompass any and all possible combinations of one or more of the associated listed items. As used herein, the phrase “at least one of” preceding a series of items, with the term “and” or “or” to separate any of the

items, modifies the list as a whole, rather than each member of the list (i.e., each item) The phrase “at least one of” does not require selection of at least one of each item listed; rather, the phrase allows a meaning that includes at least one of any one of the items, and/or at least one of any combination of the items, and/or at least one of each of the items By way of example, the phrases “at least one of A, B, and C” or “at least one of A, B, or C” may each refer to only A, only B, or only C; any combination of A, B, and C; and/or at least one of each of A, B, and C. The terms “includes,” “including,” “comprises,” and/or “comprising,” when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

[0098] The term “if” may, optionally, be construed to mean “when” or “upon” or “in response to determining” or “in response to detecting,” depending on the context. Similarly, the phrase “if it is determined” or “if [a stated condition or event] is detected” may, optionally, be construed to mean “upon determining” or “in response to determining” or “upon detecting [the stated condition or event]” or “in response to detecting [the stated condition or event],” depending on the context.

[0099] As used herein, the terms “computer,” “personal computer,” “device,” “computing device,” “router device,” and “controller device” may refer to any programmable computer system that is known or that will be developed in the future. In certain embodiments, a computer may be coupled to a network, such as described herein. A computer system may be configured with processor-executable software instructions to perform the processes described herein. Such computing devices may be mobile devices, such as a mobile telephone, data assistant, tablet computer, or other such mobile device. Alternatively, such computing devices may not be mobile (e.g., in at least certain use cases), such as in the case of server computers, desktop computing systems, or systems integrated with non-mobile components.

[0100] As used herein, the terms “component,” “module,” and “system” may be intended to refer to a computer-related entity, either hardware, a combination of hardware and software, software, or software in execution. For example, a component may be, but is not limited to being, a process running on a processor, a processor, an object, an executable, a thread of execution, a program, and/or a computer. By way of illustration, both an application running on a server and the server may be a component. One or more components may reside within a process and/or thread of execution and a component may be localized on one computer and/or distributed between two or more computers.

[0101] The predicate words “configured to,” “operable to,” and “programmed to” do not imply any particular tangible or intangible modification of a subject, but, rather, are intended to be used interchangeably. In one or more implementations, a processor configured to monitor and control an operation or a component may also mean the processor being programmed to monitor and control the operation or the processor being operable to monitor and control the operation. Likewise, a processor configured to execute code can be construed as a processor programmed to execute code or operable to execute code.

[0102] Phrases such as an aspect, the aspect, another aspect, some aspects, one or more aspects, an implementation, the implementation, another implementation, some implementations, one or more implementations, an embodiment, the embodiment, another embodiment, some implementations, one or more implementations, a configuration, the configuration, another configuration, some configurations, one or more configurations, the subject technology, the disclosure, the present disclosure, other variations thereof and alike are for convenience and do not imply that a disclosure relating to such phrase(s) is essential to the subject technology or that such disclosure applies to all configurations of the subject technology. A disclosure relating to such phrase(s) may apply to all configurations, or one or more configurations. A disclosure relating to such phrase(s) may provide one or more examples. A phrase such as an aspect or some aspects may refer to one or more aspects and vice versa, and this applies similarly to other foregoing phrases.

[0103] The word “exemplary” is used herein to mean “serving as an example, instance, or illustration”. Any embodiment described herein as “exemplary” or as an “example” is not necessarily to be construed as preferred or advantageous over other implementations. Furthermore, to the extent that the term “include,” “have,” or the like is used in the description or the claims, such term is intended to be inclusive in a manner similar to the term “comprise” as “comprise” is interpreted when employed as a transitional word in a claim.

[0104] All structural and functional equivalents to the elements of the various aspects described throughout this disclosure that are known or later come to be known to those of ordinary skill in the art are expressly incorporated herein by reference and are intended to be encompassed by the claims. Moreover, nothing disclosed herein is intended to be dedicated to the public regardless of whether such disclosure is explicitly recited in the claims. No claim element is to be construed under the provisions of 35 U.S.C. § 112, sixth paragraph, unless the element is expressly recited using the phrase “means for” or, in the case of a method claim, the element is recited using the phrase “step for”.

[0105] The previous description is provided to enable any person skilled in the art to practice the various aspects described herein. Various modifications to these aspects will be readily apparent to those skilled in the art, and the generic principles defined herein may be applied to other aspects. Thus, the claims are not intended to be limited to the aspects shown herein, but are to be accorded the full scope consistent with the language claims, wherein reference to an element in the singular is not intended to mean “one and only one” unless specifically so stated, but rather “one or more”. Unless specifically stated otherwise, the term “some” refers to one or more. Pronouns in the masculine (e.g., his) include the feminine and neuter/neutral gender (e.g., her and its and they) and vice versa. Headings and subheadings, if any, are used for convenience only and do not limit the subject disclosure.

[0106] While there have been described adjustable storage bins for vehicles and methods for using the same, many changes may be made therein without departing from the spirit and scope of the subject matter described herein in any way. Insubstantial changes from the claimed subject matter as viewed by a person with ordinary skill in the art, now known or later devised, are expressly contemplated as being equivalently within the scope of the claims. Therefore, obvious substitutions now or later known to one with ordinary skill in the art are defined to be within the scope of the defined elements. It is also to be understood that various directional and orientational terms, such as “left” and “right,” “up” and “down,” “front” and “back” and “rear,” “top” and “bottom” and “side,” “above” and “below,” “length” and “width” and “thickness” and “diameter” and “cross-section” and “longitudinal,” “X-” and “Y-” and “Z-,” and/or the like, may be used herein only for convenience, and that no fixed or absolute directional or orientational limitations are intended by the use of these terms. For example, the components of the apparatus can have any desired orientation. If reoriented, different directional or orientational terms may need to be used in their description, but that will not alter their fundamental nature as within the scope and spirit of the disclosure.

[0107] Therefore, those skilled in the art will appreciate that the concepts of the disclosure can be practiced by other than the described embodiments, which are presented for purposes of illustration rather than of limitation.

Claims

1. A vehicle comprising: a vehicle bin component defining a vehicle bin component space; and a storage bin system comprising: a bin carrier defining a carrier space, wherein the bin carrier is at least partially positioned within the vehicle bin component space; a tray; and a tray movement assembly operative to control movement within the carrier space of the tray between a plurality of tray positions relative to the bin carrier.

2. The vehicle of claim 1, wherein: the vehicle further comprises an electronic vehicle subsystem; the storage bin system further comprises a device module; the device module is physically coupled to the tray; the device module is electrically coupled to the electronic vehicle subsystem; and the device module is operative to electrically couple to a user electronic device when the user electronic device is positioned on the tray.
3. The vehicle of claim 2, wherein the device module comprises a wireless charging module.
4. The vehicle of claim 2, wherein the device module comprises a Universal Serial Bus (“USB”) connector.
5. The vehicle of claim 1, wherein the vehicle bin component comprises a center console.
6. The vehicle of claim 1, wherein the vehicle bin component comprises one of the following: a glove box; an interior storage component; a door trim; an overhead storage; a trunk trim quarter garnish; a floor panel; a seat; an upper panel; or an instrument panel storage.
7. The vehicle of claim 1, wherein the vehicle is one of the following: a road vehicle; a rail vehicle; a watercraft; or an aircraft.
8. The vehicle of claim 1, wherein the tray movement assembly comprises a push-push latch assembly.
9. The vehicle of claim 1, wherein the tray movement assembly comprises a locking multi-step assembly.
10. The vehicle of claim 1, wherein the tray movement assembly comprises a scissor-type assembly.
11. The vehicle of claim 1, wherein the tray movement assembly comprises a storage bin motor assembly.
12. The vehicle of claim 1, wherein: the vehicle bin component space comprises a bin space portion and a remaining space portion; the bin carrier is at least partially positioned within the bin space portion; the bin carrier comprises a plurality of carrier walls that at least partially defines the carrier space; a side storage bin opening extends through a carrier wall of the plurality of carrier walls; and the side storage bin opening couples the carrier space with the remaining space portion.
13. The vehicle of claim 12, wherein a tray position of the plurality of tray positions is aligned with the side storage bin opening.
14. The vehicle of claim 12, wherein the storage bin system further comprises an ejection assembly configured to eject at least one of the following from the carrier space and into the remaining space portion via the side storage bin opening once the tray is at the tray position: the tray; or an object positioned on the tray.
15. The vehicle of claim 12, wherein: the vehicle bin component space comprises a bin space portion and a remaining space portion; the bin carrier is at least partially positioned within the bin space portion; the vehicle bin component comprises a plurality of vehicle bin component walls that at least partially defines the vehicle bin component space; a vehicle bin component opening extends through a vehicle bin component wall of the plurality of vehicle bin component walls; and the vehicle bin component opening couples the remaining space portion with an ambient space that is external to the vehicle bin component space.
16. The vehicle of claim 15, wherein the vehicle further comprises: a door; and a door movement assembly operative to control movement of the door between a first door position within the vehicle bin component opening and a second door position out from the vehicle bin component opening.
17. The vehicle of claim 12, wherein the storage bin system further comprises: a door; and a door movement assembly operative to control movement of the door between a first door position within the side storage bin opening and a second door position out from the side storage bin opening.
18. The vehicle of claim 17, wherein, when the door is in the second door position, a portion of the door is positioned within the carrier space and is operative to support a user object.
19. A vehicle comprising: a vehicle bin component defining a vehicle bin component space; and a

storage bin system comprising: a bin carrier defining a carrier space, wherein the bin carrier is at least partially positioned within the vehicle bin component space; a tray; and a tray movement assembly operative to: move the tray within the carrier space along a first axis; and move the tray out from the carrier space along a second axis that is perpendicular to the first axis.

20. A vehicle comprising: a vehicle bin component defining a vehicle bin component space; and a storage bin system comprising: a bin carrier defining a carrier space, wherein the bin carrier is at least partially positioned within the vehicle bin component space; a tray; and a tray movement assembly operative to control movement within the carrier space along a first axis of the tray between a first tray position and a second tray position, wherein the bin carrier comprises a wall with an opening for enabling a user to move the tray along a second axis perpendicular to the first axis from the second tray position to a third tray position external from the carrier space.
